# Contents

HW 1 1

HW 2 6

HW 3 9

HW 4 13

HW 5 16

## Note

All of these exercises come from Utts and Heckard [1].

# HW 1

*Remark.* These solutions have roughly the level of writing that I expect to see in your homework. Notice that I have used complete and concise sentences to communicate not only the answers, but a reminder of what was asked. In cases where it's warranted, I give explanations for my answers. These explanations are just as important as the answers themselves.

### 2.3

a) An instructor surveying all the students in her class to determine their preference is gathering data from the **entire population** that she wishes to study.

b) A poll of 1000 people that is used to estimate a nationwide opinion on POTUS's job performance represents **sample data**, since in this case the population consists of all the nation's people (a set far exceeding 1000 in size).

### 2.5

a) In a census, every member of the population is counted. Therefore, the 2000 census's estimate that the average US household size was 2.59 persons represents a **population parameter**.

b) An ABC News poll finds that 36% of a 500-person sample of Americans supports a particular view. Since the goal is to estimate the percentage of Americans (the "population") in support of that view, the 36% is a **sample statistic**.

c) "To estimate average normal body temperature of all adults, a doctor measures the temperatures of 100 healthy adults. The average temperature for that group is 98.2 degrees Fahrenheit." ([1, Page 55]). Here, 98.2 is a **sample statistic**, since it is calculated from sample data, and is used to estimate a population parameter.

## 2.11

a) Importance of religion to respondent (very, somewhat, or not very important): **Categorical**. The categories are the three listed in parentheses.

b) Hours of sleep last night. **Quantitative**. One must measure this numerical quantity with a timer.

c) Weights of adult women, measured in pounds. **Quantitative**. Again, it's a numerical quantity which must me measured.

d) Favorite color for an automobile. **Categorical**. The categories are the various colors.

## 2.21

A physiologist records the pulse rates of 30 men and 30 women.

a) The two variables here are **sex** and **pulse rate**.

b) Sex is a **categorical** variable, its values being non-numerical labels. Pulse rate is a **quantitative** variable, it being a number which must be measured.

c) In order to compare pulse rates of men and women, one might ask "Do men and women have the same pulse rates on average?" It would then be useful to compute the average pulse rate $\bar{x}$ found for the 30 men, and the average pulse rate $\bar{y}$ found for the 30 women. One could then compare $\bar{x}$ and $\bar{y}$.

## 2.22

Most of the questions in the surveys students fill out at the end of a semester are ordinal. For example, *Please rate the overall quality of the course: Poor, Fair, Good, Outstanding, Exceptional*. Here, there are five categories, and they have a natural ranking from worst to best. But it would not be meaningful to average the results of the students' answers.

## 2.25

a) In order to answer the question *Is the average IQ of left-handed people higher than the average IQ of right-handed people?*, researchers should gather data for the variables of **IQ** and **handedness**. IQ is a quantitative variable whose values are a numerical score on an IQ test. Handedness is a categorical variable whose values are "right-handed"

and "left-handed." (The question, as posed, disregards the existence of the category "ambidextrous.") The question views handedness as the explanatory variable, and IQ as the response variable.

b) In the question *For married couples, is there a relationship between owning a pet and whether or not they get divorced?*, the two variables of interest are the categorical variables of **pet ownership** (with categories pet-owner and non-pet-owner) and **divorce** (with categories they-get-divorced and they-stay-together). Pet ownership is the explanatory variable, and divorce is the response variable.

## 2.30

With reference to a study comparing the grades of 12th grade students and their responses to the question of how often they wear a seatbelt (see the table on [1, Page 57]):

a) Of the 2470 students in the table

$$\frac{1700}{2470} \times 100 \approx 68.8\%$$

said that they usually get A's and B's.

b) Of the 1700 students who said that they usually get A's and B's,

$$\frac{1056}{1700} \times 100 \approx 62.1\%$$

said that they always wear a seatbelt.

c) Of the 657 students who said that they usually get C's,

$$\frac{300}{657} \times 100 \approx 45.7\%$$

said that they always wear a seatbelt.

d) Of the 113 students who said that they usually get D's and F's,

$$\frac{41}{113} \times 100 \approx 36.3\%$$

said that they always wear a seatbelt.

## 2.35

Of 92 students who responded to "Randomly pick a letter—S or Q," 61 picked S and 31 picked Q. The order of the letter choices was reversed for another 98 students who responded to "Randomly pick a letter—Q or S." Of these 98 students, 45 picked S and 53 picked Q.

a) The following two-way table contains the data and summarizes the relationship between the letter listed first and the letter picked.

| | S | Q | Total |
|---|---|---|---|
| "...S or Q." | 61 | 31 | 92 |
| "...Q or S." | 45 | 53 | 98 |
| Total | 106 | 84 | 190 |

b) Of the 92 students who saw S listed first, **66.3%** (= 61/92) picked S and the remaining **33.7%** picked Q.

c) Of the 98 students who saw Q listed first, **45.9%** (= 45/98) picked S and the remaining **54.1%** picked Q.

d) `bar chart omitted`

e) The data suggest that the order in which the letters are listed does influence the letter that is picked.

...But how strong is the suggestion? Of the 190 students, 114, or 60%, picked the first letter. If the students were truly picking randomly (with no external influence, neither from the listing order nor anything else), we would expect something close to a 50-50 split. Is 60% lopsided enough to say that the listing order had an effect? Or does it still count as being "close to a 50-50 split"? And what to make of the fact that the bias toward the first letter was more pronounced when it was S that was listed first?

These questions are not easy to answer intuitively. We will study methods for quantifying our confidence in our answers to them. In the meantime, think about how your intuition would change if similar proportions came from sample sizes that were either much smaller or much larger. Consider the hypothetical tables

| | S | Q | Total |
|---|---|---|---|
| "...S or Q." | 6 | 3 | 9 |
| "...Q or S." | 5 | 5 | 10 |
| Total | 11 | 8 | 19 |

and

| | S | Q | Total |
|---|---|---|---|
| "...S or Q." | 61000000 | 31000000 | 92000000 |
| "...Q or S." | 45000000 | 53000000 | 98000000 |
| Total | 106000000 | 84000000 | 190000000 |

The first one would convince me of nothing, while the second one would have me believing not only that there indeed existed a first-letter bias, but also an S-bias.

## 2.59

We are asked for means and medians of lists of values.

a) $\{64, 68, 72, 76, 80, 86\}$

$$\text{mean} = \frac{64 + 68 + 72 + 76 + 80 + 86}{6} = \frac{446}{6} \approx \mathbf{74.3}$$
$$\text{median} = \frac{72 + 76}{2} = \mathbf{74}$$

b) $\{10, 6, 2, 7, 100\}$

$$\text{mean} = \frac{10 + 6 + 2 + 7 + 100}{5} = \frac{125}{5} = \mathbf{25}$$
$$\text{median} = \mathbf{7}$$

c) $\{30, 10, 40, 30\}$

$$\text{mean} = \frac{30 + 10 + 40 + 30}{4} = \frac{110}{4} = \mathbf{27.5}$$
$$\text{median} = \mathbf{30}$$

## 2.74

We are given a list of point differences in a series of 11 football games:

$$+38, -14, +24, -13, -9, -7, -2, -11, -7, +4, +24$$

a) The value of the mean point difference and the value of the median point difference for the 11 games are **2.45** and $\mathbf{-7}$, respectively.

b) Of those two summary values, **the median**, $-7$, does a better job of describing the season, since (I think) what really matters is whether a team wins or loses—not necessarily the size of the margin of the loss/victory. The fact that the median is a negative number here reflects that the team lost more games than it won. By contrast, the mean, 2.45, may be telling us that when the team won, it tended to win by a lot, and when it lost, it tended to lose by a smaller margin. Perhaps that serves as some consolation to the players after a losing season.

# HW 2

## 2.37

With reference to a study where men and women were asked what the fastest speed they'd ever driven was (see the table on [1, Page 58]):

a) The fastest speed driven by anyone in the class was **150 mph**, as seen in the 'extremes' row of the five-number summary for men.

b) The slowest of the "fastest speeds" driven by a male is found in that same row, and it is **55 mph**.

c) The speed for which one-fourth of the women had driven at that speed or faster is the third quartile for the women. From the table, it is **95 mph**.

d) The proportion of females who had driven 89 mph or faster is **one half**, since 89 is the median in the females summary.

e) The number of females who had driven 89 mph or faster is **51**, since 89 is the median in the females summary and $102 = 2 \times 51$ females were sampled.

## 2.66

We are given the following set of systolic blood pressures:

$$110, 123, 132, 150, 127, 118, 102, 122.$$

a) The median for this data set is the average of 122 and 123, namely, **122.5**.

b) The lower quartile is the median of the lower half of the data points, in this case **114**, the average of 110 and 118.

c) Similarly, we find that $Q_3$ is 129.5, therefore the IQR is **15.5**.

d) Note that 1.5 times IQR is 23.25. If we use the criterion that an outlier is further from the middle quartiles than 23.25, then **there are no outliers** here.

e) `boxplot omitted`

## 2.94

With reference to the data set

$$110, 123, 132, 150, 127, 118, 102, 122$$

of systolic blood pressures:

a) The mean for these data is

$$\frac{1}{8}(110 + 123 + 132 + 150 + 127 + 118 + 102 + 122) = \frac{984}{8} = \mathbf{123}$$

and the standard deviation is **14.45**, obtained by taking the square root of the variance, found in part (b).

b) The variance for these data is

$$\frac{1}{7}\left(13^2 + 0^2 + 9^2 + 27^2 + 4^2 + 5^2 + 21^2 + 1^2\right) = \frac{1462}{7} = \mathbf{208.86}$$

## 2.96

If my score on an exam was 80 and the mean was 70, then I would be happier with a standard deviation of 5 than I would be if the standard deviation was 15. Standard deviation is a measure of the spread of the data. A smaller spread would indicate that the data were more tightly clustered around the mean of 70, making my 80 more of an achievement.

## 3.2

a) Miles of running per week and time for a 5-kilometer run are likely to be **positively associated** variables. I would expect avid runners to be faster than non-runners.

b) Forearm length and foot length are likely to be **positively associated**. Both are usually larger in taller people.

c) Grade level and height for children in grades 1 through 10 are likely to be **positively associated**. The higher the grade, the older the child; the older the child, the taller child.

## 3.18

We are given regression equation $\hat{y} = 113.6 - 1.01x$, where $y$ denotes average August temperatures and $x$ denotes geographic latitudes for 20 US cities.

a) The slope of the regression line is $\mathbf{-1.01}$. This means that a positive one-unit change in latitude is predicted to result in a decrease in temperature by 1.01 degrees.

b) At latitude $x = 32$, the regression line predicts average August temperature $\hat{y} = 81.28$ degrees.

c) At latitude $x = 38$, the regression line predicts average August temperature $\hat{y} = 75.22$ degrees. San Francisco is at latitude 38 and its mean August temperature is 64. The prediction error is $64 - 75.22 = \mathbf{-11.22}$ degrees.

## 3.22

With reference to the regression equation

$$\underbrace{\text{Pulse after marching}}_{y} = 17.8 + 0.894\,(\underbrace{\text{Resting pulse}}_{x})$$

a) The slope of this equation is 0.894. This means that if two people's resting pulse rates differ by 1 beat per minute, then it is predicted that their pulse rates after marching will differ by 0.894 beats per minute.

b) The regression predicts that a person whose resting pulse rate is 50 beats per minute will have a pulse rate of **62.5 beats per minute after marching.**

c) The regression predicts that a person whose resting pulse rate is 90 beats per minute will have a pulse rate of **98.26 beats per minute after marching.**

d) `drawing omitted`

## 3.28

a) Of the numbers

$$\mathbf{0}, -0.25, \mathbf{0.3}, \mathbf{1.0}, 1.7, \mathbf{25}\%, -50\%, +200\%,$$

only the ones in **bold font** can be values of the squared correlation coefficient, $r^2$. This is because we must always have $0 \leqslant r^2 \leqslant 1$, or, as a percentage, $0\% \leqslant r^2 \leqslant 100\%$.

b) Of the legitimate values in the previous part, the strongest linear relationship between $x$ and $y$ is represented by **1.0**. After all, the bigger the $r^2$, the stronger the relationship.

## 3.34

Two variables can have a perfect curved relationship yet have zero correlation. For example, the data may consist of the points

$$(-2, 0), \quad (-1, 3), \quad (1, 3), \quad (2, 0).$$

They lie on the parabola $y = -x^2 + 4$, a perfect curved relationship.

Meanwhile, $\bar{x} = 0$ and $\bar{y} = 1.5$. Now note that in the computation

$$r = \frac{1}{3}\sum\left(\frac{x_i - \bar{x}}{\sigma_x}\right)\left(\frac{y_i - \bar{y}}{\sigma_y}\right)$$

we find the factor

$$\begin{aligned}\sum(x_i - \bar{x})(y_i - \bar{y}) &= (-2-0)(0-1.5) + (-1-0)(3-1.5) + (1-0)(3-1.5) + (2-0)(0-1.5)\\ &= (2)(1.5) + (-1)(1.5) + (1)(1.5) + (2)(-1.5)\\ &= \mathbf{0}.\end{aligned}$$

`Sketch omitted.`

### 3.38

We are asked to assign the following correlation coefficients to the figures on [1, Page 104].

$$\underbrace{-0.9}_{\text{Graph 2}}, \qquad \underbrace{0}_{\text{Graph 3}}, \qquad \underbrace{+0.3}_{\text{Graph 4}}, \qquad \underbrace{+0.6}_{\text{Graph 1}}.$$

Only Graph 2 displays a strong negative association. The symmetry in Graph 3 makes it clear that the regression line is horizontal. Of Graphs 1 and 4, Graph 1 has a visibly stronger positive association.

### 3.44

The correlation between latitude and average August temperature (in degrees Fahrenheit) is $-0.78$ for the 20 cities shown in the table for Exercise 3.9. (The data also are in the dataset temperature on the companion website.)

a) For this data we have $\mathbf{r^2 = 0.917}$. This means that 91.7% of a city's temperature's deviation from the average is explained by that city's latitude.

b) If temperature were to be converted to Centigrade (without rounding off) the **correlation coefficient would not change**.

## HW 3

### 3.47

In a regression analysis, the total sum of squares (SSTO) is 800, and the error sum of squares (SSE) is 200. From this, we may calculate

$$r^2 = \frac{\text{SSR}}{\text{SSTO}} = \frac{\text{SSTO} - \text{SSE}}{\text{SSTO}} = \frac{800 - 200}{800} = 0.75.$$

### 3.72

It is said that a higher proportion of drivers of red cars are given tickets for traffic violations than drivers of any other car color. This **does not necessarily mean** that if you drive a red car rather than a car of some other color, it will cause you to get more tickets for traffic violations. There could be some other mutual "cause" of traffic violations and redness of car—a confounding variable. Perhaps those who feel a *need for speed* are disproportionately attracted to red cars, resulting in a disproportionately high number of speeders among drivers of red cars. In this case the *need for speed* is causing both the redness of the car and the traffic violations.

## 3.81

We are referred to a data set where $y =$ foot length (cm) and $x =$ height (in). An outlier is excluded, and the remaining data leads to regression equation $\hat{y} = 0.25 + 0.384x$.

a) The equation has slope 0.384, meaning that for each unit increase in height, there is 0.384-unit increase in foot length. Since the units are inches and centimeters (respectively), the average foot length incease for each 1-inch increase in height is **0.384 centimeters**.

b) Similarly, a 10-inch difference in height is predicted to be reflected in a **3.84 centimeter** difference in foot length.

c) Suppose Max is 70 inches tall. Then the regression equation predicts that his foot length is $0.25 + 0.384 \cdot 70 =$ **27.13 centimeters**. If his actual foot length is 28.5 centimeters, then the residual is $28.5 - 27.13 =$ **1.37 centimeters**.

## 3.82

We are given regression equation

$$\text{Winning time} = 272.63 - 0.1184(\text{Year})$$

for winning times in the Olympic men's 500-meter skating race.

a) Since the slope of the regression line is negative, the **correlation between winning time and year is negative**. That is, **as time passes, the regression line predicts that winning times decrease** (i.e., the skaters are getting faster).

b) The regression equation predicts the 2010 winning time to be $272.63 - 0.1184 \times 2010 =$ **34.646 seconds**. The actual winning time was 34.91 seconds, a **residual of 0.264 seconds**.

c) The slope $-0.1184$ tells us that we should expect winning times to decrease by as many seconds with every passing year. Since Olympics happen every 4 years, the slope tells us that from one Olympic games to the next, it is predicted that the winning time will **decrease by** $0.1184 \times 4 =$ **0.4736 seconds**.

d) We should not use this same regression line to predict the results of the 2080 Olympics. This would be an **extrapolation** far **beyond the range of the observed data**. If the regression line were continued for all future time, then eventually it would predict winning times that were negative, which is absurd. Even in the nearer future (like 2080), it may be predicting winning times that are superhuman.

| age grp | every day | few/wk | 1/wk | < 1/wk | total |
|---|---|---|---|---|---|
| 18-29 | 45 (19.23%) | 68 (29.06%) | 38 (16.24%) | 83 (35.47%) | 234 (100%) |
| 30-49 | 118 (22.78%) | 125 (24.13%) | 100 (19.31%) | 175 (33.78%) | 518 (100%) |
| $\geqslant 50$ | 260 (46.93%) | 99 (17.87%) | 69 (12.45%) | 126 (22.74%) | 554 (100%) |
| Total | 423 (32.39%) | 292 (22.36%) | 207 (15.85%) | 384 (29.40%) | 1306 (100%) |

Table 1: For 4.4(a)

## 4.4

a) We are asked to include the row percentages in Table 1.

b) From Table 1, it seems that age and frequency of reading newspapers are positively related. Among the older respondents, a higher percentage reported reading frequently.

## 4.28

*Suppose a newspaper article states that drinking three or more cups of coffee per day doubles the risk of gall bladder cancer. Before giving up coffee, what questions should be asked by a person who drinks this much coffee?*

I would ask: How was this conclusion reached? Was a study conducted? Data gathered? How were the data gathered? In short, does this result come from properly conducted research?

And, if so, I would want to know more about the data. What is the baseline risk? If the risk that is being doubled is extremely small, then I might not care enough to stop drinking coffee.

## 5.3

a) The heights of women in a psychology class are to be used to estimate the average height of all women at a college. It is of course possible (and maybe even likely) that the estimate coming from this sample is near the population parameter. Nevertheless, best practices dictate that one should sample the women at the college *randomly* in order to meaningfully estimate the average height of women at the college. The women in the psychology class constitute a convenience sample. As far-fetched as it may sound, it is in principle possible that there is some confounding variable with a non-trivial association both to height and to interest in psychology.

b) Parents of children attending a daycare are surveyed about whether the state government should increase funding to daycare providers to provide lower rates for low-income families, and these data are to be used to estimate statewide support for the increase. The possibility of selection bias is easier to conceive here than in part (a). Parents of children in daycare will naturally value the increased funding more

highly than nonparents will, resulting in data that overestimate (probably by a lot!) the statewide approval of the increased funding measure.

## 5.13

A friend has recommended the work of a musician who has recorded *five* CDs, each containing *ten* selections. You decide you will sample the musician's songs. You go to a music store and randomly choose *four* of the CDs, and then randomly choose one song from each one. **This is not a simple random sample of the musician's songs.** In a simple random sample, every possible selection of 4 of the musician's songs must be equally likely. But in this sampling method (cluster), there are some groupings of 4 songs that are altogether impossible, for example, the first four tracks on any one of the CDs.

## 5.16

We are asked to discuss which kinds of biases would be introduced in the following situations.

a) In a college town, students are hired to conduct door-to-door interviews to determine whether city residents think there should be a law forbidding loud music at parties. Since the question is about city residents' opinions, it makes a certain amount of sense to go door-to-door…to the doors of the residences where the residents reside. However, if the survey is done during the day, several residents will not be home to answer the door because they're at work. The door-to-door strategy has introduced **selection bias**. Having a day-time work schedule is probably correlated with opinion on late-night party music volume. I can also imagine some **response** bias creeping in: those who do answer the door may be reluctant to tell the students their true feelings on loud party music, out of politeness.

b) A magazine sends a survey to a random sample of its subscribers asking them whether they would like the frequency of publication reduced from biweekly to monthly or would prefer that it remain the same. There will be **nonresponse bias** here. Some of the selected participants will not mail in the survey, so the data will likely be biased away from the "I don't care" and the "happy to leave things as they are" opinions.

c) A random sample of registered voters is contacted by phone and asked whether or not they are going to vote in the upcoming presidential election. Assuming the question that is being investigated is phrased specifically with respect to *registered voters with telephones*, I see the potential for **response bias**. Some participants may prefer to express their true political stances only in the privacy of the voting booth, while telling the pollster what they think he wants to hear.

# HW 4

## 6.4

Suppose that a statistics teacher wants to know whether the number of hours students spend studying in a group affects the final course grade. In each part, explain whether the research method described is a randomized experiment or an observational study.

a) Each student keeps a log of the hours he or she spends studying in a group and reports the total hours after the course is completed. **Observational study.** There is no randomization, no treatment.

b) Students are randomly assigned to study groups. The teacher tells each group how often to meet. This varies from 1 hour the day before each exam to 2 hours per week. **Randomized experiment** with the treatment being study duration.

c) Students voluntarily join groups based on how often the groups will meet. The groups are designated as meeting weekly, meeting only before exams, or meeting when- ever enough members feel that it is necessary.

## 6.5

Remember that a confounding variable affects the response variable and is related to the explanatory variable. For each of the following situations, explain how the given confounding variable meets these criteria.

a) *Response variable*: Math skills of children aged 6 to 12 years old; *explanatory variable*: shoe size; *confounding variable*: age.

   Age and shoe size are positively associated in children aged 6 to 12. And age affects a child's math skills. Both increase together.

b) *Response variable*: Whether or not a student gets a cold during a school term; *explanatory variable*: whether or not the student procrastinates in doing assignments; *confounding variable*: number of hours per week the student spends socializing.

   Those who spend more time socializing are more likely to procrastinate on their assignments. They are also more likely to catch a cold, since they are exposed to more people.

## 6.14

A news article on the Reuters Health website, December 18, 1998, reported that:

> A study of over 1200 people over age 65 showed that the "owls"—those who go to sleep after 11 p.m. and rise after 8 a.m.—tend to be as healthy and intelligent as "larks"— those who go to bed before 11 p.m. and rise before 8 a.m.— according

to a report in the December 19/26 issue of the British Medical Journal [Vol. 317, pp. 1675–1677]. The researchers also found that owls tend to have higher average income than the early birds.

a) Do you think this is based on an observational study or a randomized experiment? Explain.

b) It is mentioned that the "owls" have higher average in- come than the "larks" (early birds). Explain whether or not income is a possible confounding variable in this study.

c) Give an example of a possible confounding variable that is not mentioned.

d) Draw a figure similar to Figure 6.2 (p. 196), illustrating the steps in this study.

## 7.3

Jan is a member of a class with 20 students that meets daily. Each day for a week (Monday to Friday), a student in Jan's class is randomly selected to explain how to solve a homework problem. Once a student has been selected, he or she is not selected again that week. If Jan was not one of the four students selected earlier in the week, what is the probability that she will be picked on Friday? Explain how you found your answer.

Since Jan has not been selected earlier in the week, she is one of 16 equally likely outcomes for Friday's selection. Therefore,

$$\mathbb{P}\left(\text{Jan is picked on Friday}\right) = \frac{1}{16},$$

that is, the probability Jan is picked on Friday is 1/16.

## 7.7

Give an example of a random circumstance in which:

a) *The outcome is not determined until we observe it.*

b) *The outcome is already determined, but our knowledge of it is uncertain.*

## 7.20

A student wants to send a bouquet of roses to her mother for Mother's Day. She can afford to buy only two types of roses and decides to randomly pick two different varieties from the following four choices: Blue Bell, Yellow Success, Sahara, and Aphrodite. Label these varieties B, Y, S, A.

a) The simple events are pairs, and the sample space consists of six of them:

$$\Omega = \{(\mathrm{B}, \mathrm{Y}), (\mathrm{B}, \mathrm{S}), (\mathrm{B}, \mathrm{A}), (\mathrm{Y}, \mathrm{S}), (\mathrm{Y}, \mathrm{A}), (\mathrm{S}, \mathrm{A})\}$$

b) Assuming that all outcomes are equally likely, the probability that she will pick Sahara and Aphrodite is

$$\mathbb{P}(\{(S, A)\}) = \frac{\#\{(S, A)\}}{\#\Omega} = \frac{1}{6}.$$

Putting it another way, there are six equally likely outcomes whose probabilities must sum to 1. Any one of those outcomes therefore has probability 1/6.

## 7.22

Suppose that we randomly select a student and record how many days in the previous week the student exercised for at least half an hour. Let $\omega_i$ $(i = 0, 1, \ldots, 7)$ denote the outcome where the randomly selected student reports $i$ days of exercise.

a) The event that the student exercised on at least 5 days is $\{\omega_5, \omega_6, \omega_7\}$.

b) The event that the student exercised on at most 2 days is $\{\omega_0, \omega_1, \omega_2\}$.

c) The event that the student did not exercise at all is $\{\omega_0\}$.

## 7.26

Suppose that events A and B are mutually exclusive with $\mathbb{P}(A) = 1/2$ and $\mathbb{P}(B) = 1/3$.

a) The events A and B are **not independent**. Since they are mutually exclusive, $\mathbb{P}(A \cap B) = 0$. But $\mathbb{P}(A)\mathbb{P}(B) = 1/6 \neq 0$. Independent events must satisfy $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$.

b) The events A and B are **not complementary**. If they were, their probabilities would sum to 1. But they don't.

## 7.30

Two fair dice, one red and one green, are tossed. Explain whether the following pairs of events are mutually exclusive, independent, both, or neither:

a) The events

$$A = \{\text{red die and green die sum to 4}\} \quad \text{and} \quad B = \{\text{red die is a 3}\}$$

are **not mutually exclusive** because they both contain the simple event "red die is 3, green die is 1." They are also **not independent**. Note that $\mathbb{P}(B) = 1/6$. But $\mathbb{P}(B \mid A) \neq 1/6$. (If we know that the two dice sum to 4, the probability that the red die is a 3 is no longer 1/6. What is it?)

b) The events

$$A = \{\text{red die and green die sum to 4}\} \quad \text{and} \quad B = \{\text{red die is a 4}\}$$

are **mutually exclusive**. If the red die is 4, then the sum of the two dice must exceed 4. Since any one of the two events makes the other event impossible, they are **not independent**.

### 7.38

Two fair dice are rolled. The event

$$A = \{\text{getting the same number on both dice}\}$$

has probability 1/6.

a) In words, $A^c$ is the event that the dice land on distinct numbers.

b) The probability of that event is $\mathbb{P}(A^c) = 1 - \mathbb{P}(A) = 5/6$.

## HW 5

## References

[1] Utts and Heckard. *Mind on Statistics*. 5th Edition.